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Analysis of ECG waveform using Deep Learning architectures

1. Abstract:

Electrocardiogram (ECG) signal classification remains a crucial challenge in automated cardiac diagnosis due to the complex temporal dynamics and morphological variability of cardiac waveforms. This research presents a progressive series of three contributions addressing feature diversity, temporal dependency, and blending representation of image learning for accurate and computationally efficient ECG classification. The first contribution, CHAF (Convolution Neural Network with Heterogeneous Activation Functions), explores the role of activation diversity in enhancing the expressive power of Convolution Neural Network (CNNs). By employing distinct activation functions across convolutional layers and integrating machine learning (ML) classifiers such as K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest (RF) and Extreme Gradient Boosting (XGBoost) in place of dense layers, CHAF achieves improved classification accuracy and faster convergence on the PTB Diagnostic ECG database. The second contribution, ECG-ResViT, introduces a hybrid CNN–Vision Transformer (ViT) architecture that combines residual and dilated convolutions for local feature extraction with transformer-based self-attention for modeling long-range dependencies. This integration enables efficient representation learning with reduced computational overhead, capturing both fine-grained morphological details and global temporal relationships in ECG signals. The third contribution, BlendNet, proposes a novel image-based framework that transforms 1D ECG signals into 2D scalogram and binary images using Continuous Wavelet Transform (CWT) and thresholding. Through alpha blending, complementary spatial and morphological features are fused, followed by classification using either dense neural layers and lightweight ML algorithms. BlendNet effectively enriches ECG feature space, enhancing diagnostic precision while maintaining computational efficiency. Collectively, these three models—CHAF, ECG-ResViT, and BlendNet—form a comprehensive pipeline that advances ECG classification from activation-level optimization to hybrid temporal modeling and finally to multi-image fusion, offering improved accuracy, interpretability, and real-time adaptability in cardiac signal analysis.

2. Contribution 1: CHAF – Heterogeneous Activation Function-Based CNN for ECG Classification

2.1 Motivation and Overview

Activation functions (AFs) play a crucial role in DL architectures by introducing non-linearity and enabling models to learn complex relationships between inputs and outputs. However, using a single activation function across all layers of a CNN can lead to saturation of gradients, slower convergence, and limited representational diversity. To overcome these limitations, we propose a CNN architecture that integrates heterogeneous activation functions (CHAF) to enhance both accuracy and execution speed in ECG classification tasks.

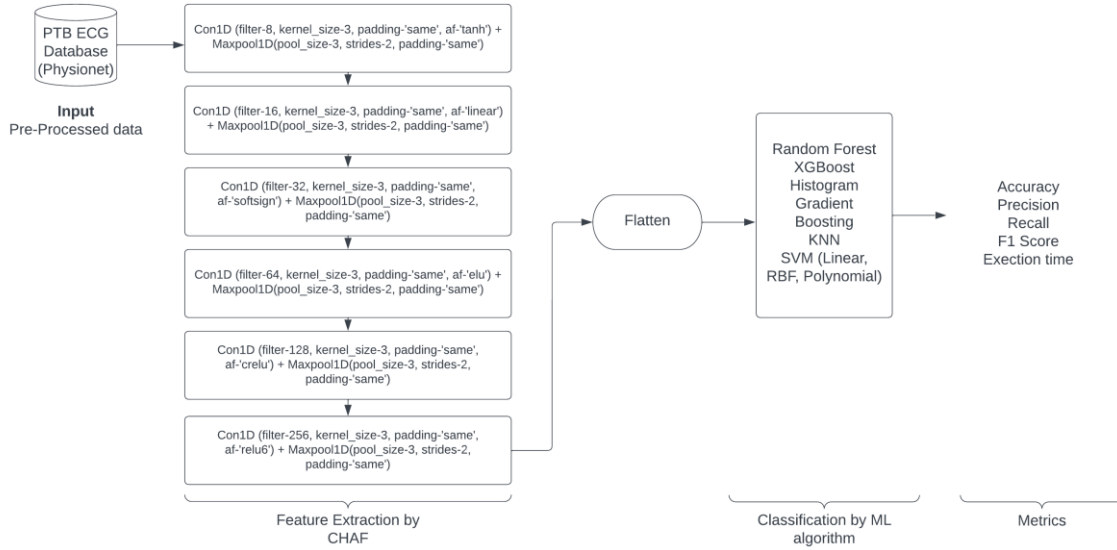


Figure 1: Architecture of CHAF Framework

2.2 Architecture and Methodology

We initially built a customized, fully connected CNN architecture at the first level and obtained good accuracy, although with longer execution times as the number of convolutional blocks increased. The model initially utilized homogeneous AF (ReLU) across all layers. However, to further improve performance, different activation functions were employed homogeneously for the same CNN structure, leading to variations in both accuracy and computational efficiency.

From these experiments, six CNN blocks were selected based on performance, and the top six AF for each block were chosen for further analysis. The architecture was subsequently modified by decoupling the feature extraction and classification stages: CNN layers performed feature extraction, while the dense layers were replaced by ML classifiers such as SVM, KNN, RF and XGBoost.

This hybridization led to enhanced accuracy and faster execution compared to the baseline CNN model. The CHAF implementation, when applied to the PTB Diagnostic ECG database, demonstrated the ability to balance feature richness with computational efficiency. Figure 1 illustrates the overall CHAF architecture integrating heterogeneous activation functions and ML classifiers.

2.3 Observations and Discussion

The CNN acted as a powerful feature extractor capable of capturing spatial and temporal dependencies in ECG signals, while the ML classifiers provided improved interpretability and reduced training time. Introducing heterogeneous AF mitigated gradient saturation issues, encouraged diversity in learned representations, and accelerated convergence.

In summary, the CHAF-based CNN-ML hybrid approach proved effective for ECG classification, providing a solid foundation for the next phase—developing architectures that not only capture local features but also model global dependencies within ECG signals.

3. Contribution 2: ECG-ResViT – Hybrid CNN-ViT Model for Efficient ECG Signal Classification

3.1 Overview

While CNNs are adept at learning local patterns, they often struggle to model long-range temporal dependencies inherent in ECG signals. To address this limitation, the ECG-ResViT model integrates the strengths of CNN-based local feature extraction and ViT-based global context modeling. The architecture combines residual blocks, dilated convolutions, and

sinusoidal positional encodings to enhance representation power while maintaining computational efficiency.

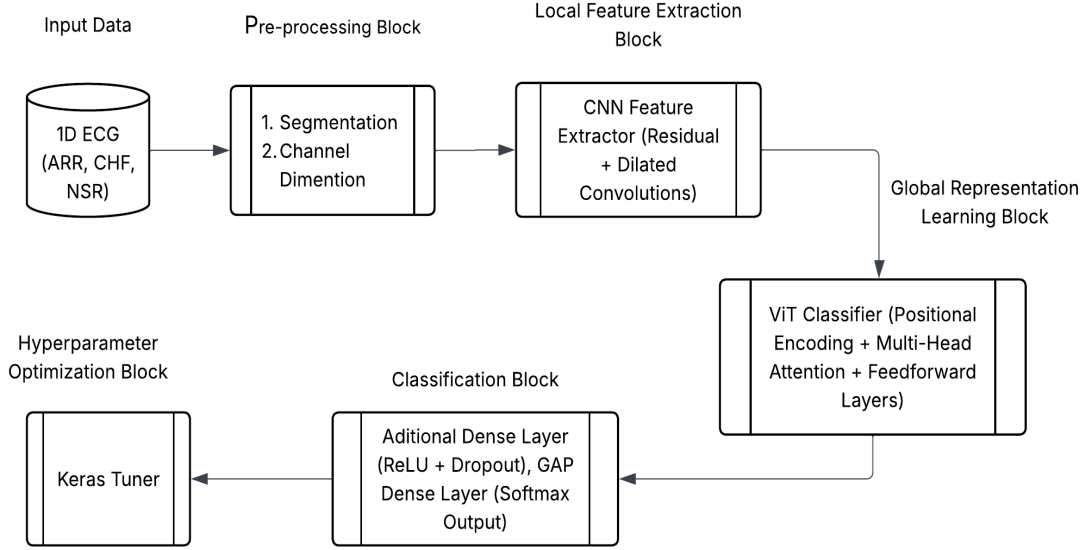


Figure 2: Proposed ECG-ResViT architecture.

3.1.1 Pre-processing Block

The ECG-ResViT pipeline begins with raw ECG signal pre-processing. ECG signals, often noisy and of variable length, are segmented into fixed-length intervals (e.g., 500 samples per segment) to maintain consistency and emphasize localized patterns. A channel dimension is added to prepare the data for Conv1D operations, enabling effective feature extraction. These steps convert unstructured 1D ECG data into structured input suitable for DL.

3.1.2 Local Feature Extraction Block

After pre-processing, the segmented ECG signals are fed into a CNN-based feature extractor. The CNN includes: **Residual Blocks:** Employ skip connections to counteract vanishing gradients, allowing deeper and more stable training. **Dilated Convolutions:** Capture multi-scale temporal features using varying dilation rates (e.g., 2, 4, 8, 16, 32), ensuring broad receptive fields without excessive parameters.

The extracted high-dimensional feature maps are then reduced using Global Average Pooling (GAP) to produce compact, fixed-length representations suitable for further processing. This step preserves essential temporal information while minimizing over fitting risks.

3.1.3 Global Representation Learning Block

Unlike traditional CNNs, ViTs can model global dependencies by attending to relationships across the entire input sequence. To adapt ViT for 1D ECG signals, several modifications were made: Sequence-to-Embedding Conversion: The ECG segments are reshaped into embeddings, where each segment functions as an input token. Positional Encodings: Sinusoidal positional encodings maintain temporal order and preserve physiological event sequences such as P-wave, QRS complex, and T-wave locations. Multi-Head Attention: Enables parallel attention to multiple temporal dependencies, allowing the network to capture both short-term and long-term patterns critical for arrhythmia detection.

3.1.4 Classification Block

The classification stage consists of: A dense layer with ReLU and dropout to learn non-linear feature interactions while preventing over fitting. A Global Average Pooling layer to summarize temporal embeddings into a single vector representation. A softmax output layer for multi-class classification (e.g., ARR, CHF, NSR). Mathematically, the softmax layer converts logits into probabilities, and the class with the highest probability is chosen as the prediction. Overall, the ECG-ResViT architecture effectively combines local feature extraction (via CNN) with global temporal modeling (via ViT), resulting in high classification accuracy with reduced computational burden. This hybridization marks a significant step toward robust and interpretable ECG analysis systems.

4. Contribution 3: BlendNet – Alpha-Blended Composite Image-Based ECG Classification

4.1 Overview

The third contribution, BlendNet, explores a novel representation learning approach by transforming 1D ECG signals into 2D image representations and blending multiple feature

domains to enhance diagnostic capability. The key hypothesis is that composite feature sets derived from different representations of the same ECG signal—specifically scalogram and binary images—capture both morphological and temporal attributes more effectively than traditional methods. The architecture comprises three main modules: (i) Image formation and binarization, (ii) Alphablending, and (iii) Classification. Figure 3 illustrates the proposed BlendNet architecture.

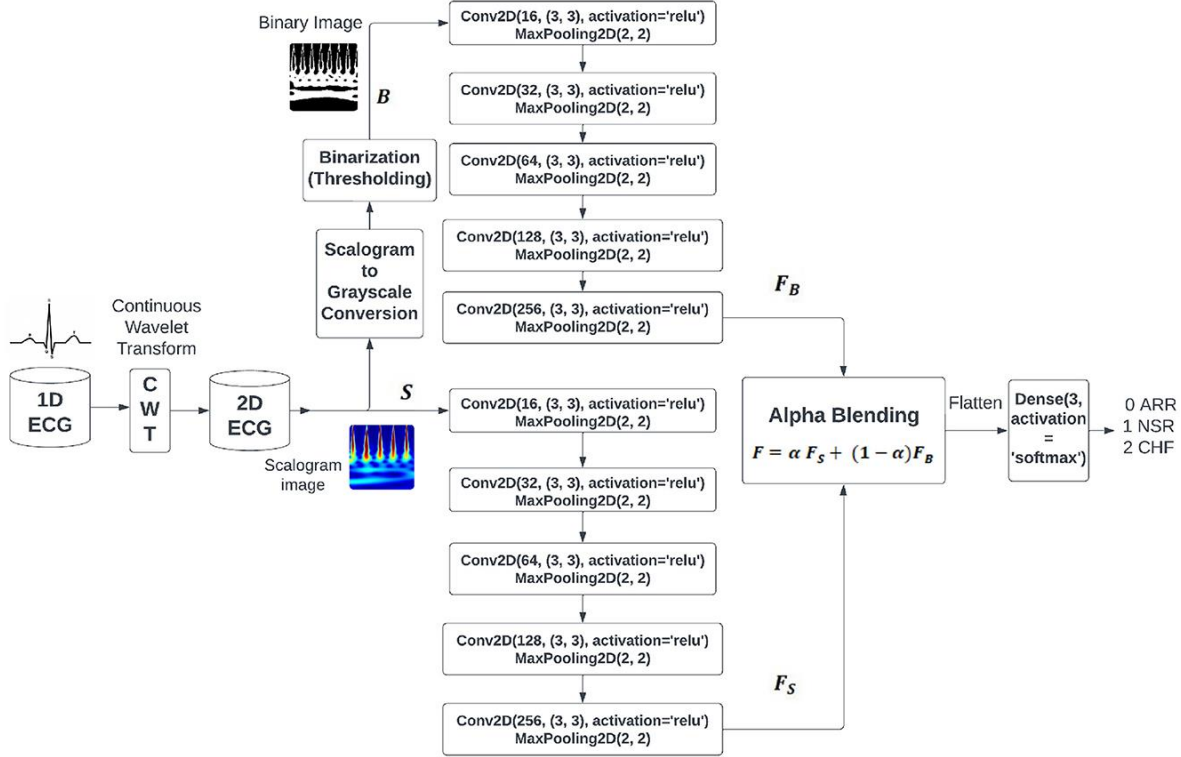


Figure 3: Proposed BlendNet Architecture

4.1.1 Image Formation and Binarization

CWT is used to convert 1D ECG signals into 2D scalogram, providing a time-frequency representation of the signal. Each point in a scalogram reflects the signal's energy distribution across time and frequency. The transformation is defined as:

$$CWT(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \Psi^* \left(\frac{t-b}{a} \right) dt$$

where a is the scale parameter, b is the translation parameter, and ψ^* is the complex conjugate of the mother wavelet.

After scalogram formation, grayscale conversion is performed, followed by binarization using thresholding (e.g., Otsu's method). This step enhances regions of interest—QRS complexes, P-waves, and T-waves—while reducing background noise.

4.1.2 Alpha Blending

Once scalogram and binary images are obtained, they are processed through parallel convolution and pooling layers using ReLU activation. The outputs of these branches are combined using alpha blending, a weighted summation technique widely applied in medical image processing:

$$F = \alpha F_S + (1 - \alpha) F_B$$

where F_S and F_B denote feature sets from scalogram and binary branches respectively, and α is the blending coefficient ($0 \leq \alpha \leq 1$).

This blended representation fuses complementary spatial and morphological features, leading to richer discriminative embeddings. The blended features are subsequently used for classification.

4.1.3 Classification Using Dense Layer

The classification layer employs a fully connected dense network with softmax activation to categorize ECG images into classes such as ARR, CHR, and NSR. The dense layer can be replaced with efficient deep architectures like InceptionV3, ResNet152V2, DenseNet169, or MobileNetV2 for enhanced performance. Alternatively, lightweight ML classifiers such as SVM, RF, KNN, or XGBoost can replace the dense layer for faster convergence in real-time applications.

5. Comparative Summary and Future Directions

Across the three contributions, this research presents a progressive exploration of ECG signal classification through different architectural perspectives:

1. CHAF (Contribution 1) focused on enhancing activation dynamics and integrating machine learning classifiers for efficient CNN-based ECG classification.
2. ECG-ResViT (Contribution 2) introduced a hybrid deep learning framework combining CNNs for local features and ViTs for global temporal dependencies, achieving improved accuracy and robustness.
3. BlendNet (Contribution 3) advanced toward multi-representation fusion by leveraging 2D image formation, alpha blending, and composite feature learning to achieve high interpretability and diagnostic performance.

Future directions include:

- Integrating Explainable AI (XAI) modules such as Grad-CAM to visualize and interpret decision regions in ECG classification models.
- Extending BlendNet to handle 3D VCG-style representations for comprehensive cardiac diagnostics.
- Exploring meta-learning and transfer learning to improve model adaptability across datasets and patient populations.

Publication details:

1. Premanand S and Narayanan S (2025) BlendNet: a blending-based convolutional neural network for effective deep learning of electrocardiogram signals. *Front. Artif. Intell.* 8:1625637. doi: 10.3389/frai.2025.1625637 (SCI Q1 – IF 4.7)
2. Premanand, S., Narayanan, S. ECG-ResViT: a hybrid CNN-ViT model for efficient ECG signal classification. *Eur. Phys. J. Spec. Top.* 234, 4483–4496 (2025). <https://doi.org/10.1140/epjs/s11734-025-01781-y> (SCI Q2 - IF 2.3)
3. P. S. and S. Narayanan, “Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals,” *Comput. Mater. Contin.*, vol. 77, no. 1, pp. 25–45, 2023. <https://doi.org/10.32604/cmc.2023.042590> (SCI Q2 - IF 3.1)
4. S. Premanand and S. Narayanan, “A Tree Based Machine Learning Approach for PTB Diagnostic Dataset,” *Journal of Physics: Conference Series*, vol. 2115, no. 1, p. 012042, Sep. 2021, doi: 10.1088/1742-6596/2115/1/012042 (Scopus)



ARTICLE

Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals

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ABSTRACT

Machine Learning (ML) and Deep Learning (DL) technologies are revolutionizing the medical domain, especially with Electrocardiogram (ECG), by providing new tools and techniques for diagnosing, treating, and preventing diseases. However, DL architectures are computationally more demanding. In recent years, researchers have focused on combining the computationally less intensive portion of the DL architectures with ML approaches, say for example, combining the convolutional layer blocks of Convolution Neural Networks (CNNs) into ML algorithms such as Extreme Gradient Boosting (XGBoost) and K-Nearest Neighbor (KNN) resulting in CNN-XGBoost and CNN-KNN, respectively. However, these approaches are homogenous in the sense that they use a fixed Activation Function (AFs) in the sequence of convolution and pooling layers, thereby limiting the ability to capture unique features. Since various AFs are readily available and each could capture unique features, we propose a Convolution-based Heterogeneous Activation Facility (CHAF) which uses multiple AFs in the convolution layer blocks, one for each block, with a motivation of extracting features in a better manner to improve the accuracy. The proposed CHAF approach is validated on PTB and shown to outperform the homogeneous approaches such as CNN-KNN and CNN-XGBoost. For PTB dataset, proposed CHAF-KNN has an accuracy of 99.55% and an F1 score of 99.68% in just 0.008 s, outperforming the state-of-the-art CNN-XGBoost which has an accuracy of 99.38% and an F1 score of 99.32% in 1.23 s. To validate the generality of the proposed CHAF, experiments were repeated on MIT-BIH dataset, and the proposed CHAF-KNN is shown to outperform CNN-KNN and CNN-XGBoost.

KEYWORDS

Electrocardiogram; convolution neural network; machine learning; activation function

1 Introduction

It is well known that many diseases affect our health in the human ecosystem. According to World Health Organization [1], there are some shocking statistics: Due to heart-related problems, around 17.9 million people die each year, which accounts for 32% of deaths globally. In this, more than 75% of death occurs in low and middle-income countries like India, Malaysia, Pakistan, Sri Lanka, Thailand, Indonesia, and many more. An Electrocardiogram (ECG) is a non-invasive medical diagnostic device that cardiologists and medical practitioners predominately use to record the heart's physiological activity throughout time.





ECG-ResViT: a hybrid CNN-ViT model for efficient ECG signal classification

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Abstract Electrocardiogram (ECG) analysis is a critical tool in the early detection of life-threatening cardiac conditions, such as Arrhythmia (ARR), Congestive Heart Failure (CHF), and Normal Sinus Rhythm (NSR). While deep learning (DL) models, especially Convolutional Neural Networks (CNNs), have demonstrated success in identifying local patterns in ECG signals, their limited capacity to capture long-range dependencies often impacts performance. On the other hand, Vision Transformers (ViTs), known for their strength in modeling global relationships, are underexplored in the context of 1D biomedical signals. This study introduces ECG-ResViT, a novel hybrid DL architecture that combines the local feature extraction power of CNNs—enhanced with residual blocks and dilated convolutions—with the global attention modeling of ViTs. The architecture leverages positional encoding and multi-head self-attention to effectively capture temporal dependencies in ECG signals. Evaluated on benchmark datasets comprising ARR, CHF, and NSR classes, the proposed model achieves an accuracy of 99.13% and ROC-AUC of 99.94%, outperforming several existing state-of-the-art methods. By balancing accuracy, interpretability, and computational efficiency, ECG-ResViT demonstrates strong potential for real-time clinical deployment and highlights the value of hybrid approaches in advancing biomedical signal classification.

1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death globally, claiming an estimated 17.9 million lives each year, which accounts for nearly 32% of all global deaths as reported by World Health Organization (WHO) [1]. Among these, timely diagnosis and classification of heart conditions like ARR, CHF, and NSR are critical in preventing severe complications. ECG signal analysis has proven to be a fundamental tool in clinical settings for monitoring heart activity and detecting abnormalities.

While experienced medical professionals are proficient at interpreting ECG signals, manual analysis can be time-consuming, subjective, and susceptible to variations, especially in emergency or real-time settings. This is where automated DL-based solutions come into play—offering speed, consistency, and the potential for real-time clinical assistance.

In recent years, DL models have gained popularity for ECG classification due to their ability to automatically learn complex patterns from raw data. CNN, in particular, are effective in detecting localized features like P waves, QRS complexes, and T waves. However, a known limitation of CNNs is their difficulty in modeling long-range temporal dependencies, which are equally important in analyzing the full scope of cardiac activity. To address this, enhancements, such as residual connections and dilated convolutions, are integrated into the CNN blocks. These techniques help the model retain important gradients during training and allow a wider receptive field, respectively—without increasing the computational load significantly.

On the other hand, Transformer-based models—originally developed for natural language tasks—have recently shown promising results in handling sequential data. ViT model, in particular, has extended this capability to vision and time-series tasks by treating input sequences as image-like patches. Its self-attention mechanism makes it effective in learning long-term dependencies, something CNNs inherently struggle with. Yet, ViTs often underperform in capturing local, fine-grained details, especially in one-dimensional signals like ECGs.

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BlendNet: a blending-based convolutional neural network for effective deep learning of electrocardiogram signals

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Introduction: In recent years, Deep Learning (DL) architectures such as Convolutional Neural Network (CNN) and its variants have been shown to be effective in the diagnosis of cardiovascular disease from ElectroCardioGram (ECG) signals. In the case of ECG as a one-dimensional signal, 1-D CNNs are deployed, whereas in the case of a 2D-represented ECG signal, i.e., two-dimensional signal, 2-D CNNs or other relevant architectures are deployed. Since 2D-represented ECG signals facilitate better feature extraction, it is a common practice to convert an ECG signal into a scalogram image using a continuous wavelet transform (CWT) approach and then subject it to a DL architecture such as 2-D CNN. However, this traditional approach captures only a limited set of features of ECG and thereby limits the effectiveness of DL architectures in disease detection.

Methods: This work proposes “BlendNet,” a DL architecture that effectively extracts the features of an ECG signal using a blending approach termed “alpha blending.” First, the 1-D ECG signal is converted into a scalogram image using CWT, and a binary version of the scalogram image is also obtained. Then, both the scalogram and binary images are subjected to a sequence of convolution and pooling layers, and the resulting feature images are blended. This blended feature image is subjected to a dense layer that classifies the image. The blending is flexible, and it is controlled by a parameter α , hence the process is termed as alpha blending. The utilization of alpha blending facilitates the generation of a composite feature set that incorporates different characteristics from both the scalogram and binary versions.

Results: For experiments, a total of 162 ECG recordings from the PhysioNet database were used. Experimental results and analysis show that, in the case of $\alpha = 0.7$, BlendNet’s performance surpasses the performance of (i) traditional approaches (that do not involve blending) and (ii) state-of-the-art approaches for ECG classification.

Discussion: Experimental outcomes show that the proposed BlendNet is flexible regarding dense layer settings and can accommodate faster alternatives [i.e., machine learning (ML) algorithms] for faster convergence. The superior performance at $\alpha = 0.7$ indicates that alpha blending allows for richer composite feature sets, leading to improved classification accuracy over conventional feature extraction and classification methods.

KEYWORDS

electrocardiogram, convolution neural network, scalogram, image blending, binary image



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OFFICE MEMORANDUM

Consequent to **Mr. PREMANAND S** (Reg.No. 19PHD1047), an Internal Full Time registrant for the Ph.D. degree in the School of Electronics Engineering successfully completing the prescribed course work for the Ph.D. programme and passing the Comprehensive Examination, his candidacy for Ph.D. degree is hereby confirmed with effect from 21.12.2023.

Dr. Sabumon P. C.
Dean- Academic Research



To
Mr. PREMANAND S (Reg.No.19PHD1047),
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